$Ti_3C_2T_x$ MXene — Single-/Few-Layer Powder

Research-grade two-dimensional transition-metal carbide powder

$Ti_3C_2T_x$ MXene is supplied as a freeze-dried single-/few-layer powder derived from Ti_3AlC_2 using a LiF/HCl-derived etching route. The material has a layer count of ≤5 and mixed −O, −OH, −F and −Cl surface terminations. This grade is supplied for electrochemical, electrocatalysis, conductive-composite and interface research.

KEY FEATURES

- ≤5 layers; freeze-dried single-/few-layer powder
- Typical conductivity of 3500 ± 500 S/cm
- Mixed −O, −OH, −F, −Cl surface terminations

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal–sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

QUALITY / DOCUMENTATION

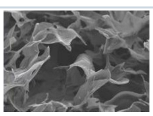
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

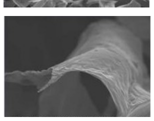
TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	$Ti_3C_2T_x$ MXene
Precursor	Ti_3AlC_2
Product form	Freeze-dried single-/few-layer powder
Preparation	LiF/HCl-derived etching route
Layer structure / count	≤5 layers
Typical conductivity	3500 ± 500 S/cm
Surface terminations	−O, −OH, −F, −Cl
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION





Single-/few-layer $Ti_3C_2T_x$ — SEM. Representative characterization; not lot-specific.

Product positioning: Supplied and technically supported by Winigen Materials for research and development use.